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Course: CSA0421 - Operating System **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 11: MASTER OS PLAN & ARCHITECTURAL DEFENSE

Comprehensive Hospital Operating System Architecture Blueprint

1. FINAL MASTER SYSTEM BLUEPRINT ARCHITECTURE

Domain	OS Type & Structure	Services & I/O Method	Scalability Strategy (3 Hospitals)
Patient Monitors	Hard RTOS (QNX Microkernel)	Minimal services, Interrupt-driven I/O, static page table protection.	Isolated VLAN edge deployment; zero cross-hospital latency dependency.
Database Server	Active-Passive Clustered Linux (LKM)	POSIX File system, Direct NVMe I/O, SELinux MAC isolation.	Multi-region database replication over dedicated fiber interconnects.
Imaging Server	SMP Linux (Modular Structure)	High-throughput VFS, DMA image transfer, POSIX threads.	Distributed GPU compute grid accepting parallel DICOM workloads.

2. ARCHITECTURAL LIMITATIONS AND RISK MITIGATION

Heterogeneous Management Overhead: Maintaining three distinct operating environments (RTOS microkernels, mobile handhelds, and clustered Linux servers) increases administrator training costs. Mitigation: Utilize unified Ansible/Puppet configuration management scripts across Linux servers and containerize workstation applications.

3. ALTERNATIVE CONCEPT & VIRTUALIZATION OVERHEAD EVALUATION

> **Alternative Scenario: Consolidated Hypervisor Virtualization (VMware/KVM on 2 Bare-Metal Servers)**
>
> Consolidating all 500 patient monitors and database workloads onto virtual machines managed by a hypervisor:
>
> - **Latency Degradation:** Hypervisor trap-and-emulate handling introduces variable scheduling jitter (`> 15 ms` latency spikes), violating hard real-time patient alarm deadlines.
> - **Hypervisor Overhead:** CPU and memory virtualization overhead consumes `8 – 12%` of total server compute resources.
> - **Single Point of Failure:** A physical motherboard or hypervisor panic brings down all virtual monitors simultaneously. Bare-metal RTOS deployment remains mandatory for patient safety.